A 5G Americas White Paper
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Enabling Intent-Based Autonomous Networks

Contents

Executive summary

Telecommunications networks are reaching an inflection point. The convergence of 5G, IoT, edge computing, and AI has driven complexity to unprecedented levels, exposing the limits of manual and rule-based automation. As the industry moves toward 6G, intent-based autonomous networks (ANs) offer the only viable path forward—systems that can self-configure, self-optimize, and self-heal while aligning operations with high-level business goals.

Globally, momentum around ANs is accelerating, with standards bodies such as TM Forum, O-RAN Alliance, and 3GPP providing foundational frameworks. Yet it is North America that is uniquely positioned to lead this transition, building on its track record of pioneering wireless innovation and its robust ecosystem of operators, vendors, and technology leaders. Early adoption of AI/Machine Learning (ML), Generative AI, and Agentic AI, combined with active engagement in standards development, provides North America with the opportunity to set the benchmark for scalable, secure, and interoperable ANs.

By spearheading high-value, near-term use cases—such as customer experience assurance, fault management, energy optimization, and dynamic resource allocation—North American operators are not merely catching up to global trends, but defining best practices that others will follow. These initial deployments establish the foundation for advancing toward higher levels of autonomy, demonstrating tangible business value while preparing networks for production-grade implementations.

The imperative is clear: North America must lead in shaping the future of telecommunications through intent-based ANs. Doing so will unlock operational efficiency, elevate customer experience, enable sustainability, and create new revenue opportunities, positioning the region as the global pioneer of the next generation of connectivity.

1. Introduction

The telecommunications industry is entering a new era defined by rising network complexity, exponential data growth, and the demand for agile and reliable connectivity. These challenges, amplified by 5G rollouts, edge computing, and AI, call for a revolutionary shift in network management practices. Intent-based ANs provide a systematic solution by mapping high-level business objectives (intents) into adaptive network actions. These capabilities collectively support efficient, resilient, and adaptive network operations aligned with evolving business objectives.

This white paper explores the key drivers, vision and practical roadmap for achieving ANs. It delves into the building blocks enabling autonomy, including intent framework, open APIs, advanced AI capabilities, observability frameworks, and Zero-Trust security, while presenting the critical role of standardization efforts led by TM Forum, O-RAN Alliance, and 3GPP. A global perspective on AN maturity, coupled with insights into North America's progress, provides a comparative lens on regional advances and challenges.

Readers will gain actionable strategies for addressing adoption hurdles—data management, governance, and interoperability—while preparing for higher autonomy levels (AN Level 4/5). A phased roadmap details practical steps, from identifying and implementing targeted, high-value/low-risk scenarios to scaling production-grade ANs. By securing industry-wide standards alignment, embracing multi-domain orchestration stakeholders can catalyze the full potential of adaptive, efficient, and scalable networks that will sustain the telecom industry for generations to come.

By aligning industry standards, fostering multi-domain orchestration, and driving collaboration across operators, vendors, and regulators, stakeholders can harness the transformative potential of adaptive, secure, and efficient networks. This coordinated effort will not only enable the telecom industry to evolve sustainably but also unlock new monetization opportunities, ensure service excellence, and create a robust foundation for future-ready networks.

2. Autonomous networks imperative

ANs were first conceptualized by TM Forum in 2019 as a revolutionary approach to network management. This concept gained momentum within the communications service provider (CSP) and vendor communities. TM Forum has developed implementation guides, architectural models, and a manifesto endorsed by hundreds of industry stakeholders, providing a foundational framework for the future of network autonomy.

At its core, the vision for ANs is creating a new generation of telecom networks that can seamlessly operate and manage themselves.

2.1 Industry drivers

Autonomous network adoption is fueled by the intersection of technological inevitability and strategic necessity. While AI and programmability form the technical foundation, the ultimate drivers lie in business agility, operational efficiency, customer experience, and sustainability. The momentum is reinforced by global ecosystem collaboration and a shared recognition that future networks must evolve from automation-driven to intent-driven and self-governing systems.

2.1.1 Rising network complexity

Telecommunication networks are becoming vast, heterogeneous, and distributed systems spanning 5G, edge, cloud, transport, enterprise, and non-terrestrial domains. Traditional operations models that rely heavily on manual interventions and rule-based automation cannot keep pace with the scale, dynamism, and cross-domain interdependencies. AN adoption is driven by the imperative to simplify management, reduce errors, and optimize performance across increasingly complex infrastructures.

2.1.2 Demand for service agility and differentiation

CSPs face mounting pressure to deliver on-demand, personalized, and context-aware services. Emerging use cases such as dynamic 5G slicing, ultra-low latency IoT, and private networks demand real-time configuration and assurance that cannot be achieved through static automation alone. AN enables intent-driven orchestration that aligns network behavior with high-level business goals, unlocking new revenue streams while ensuring differentiated customer experiences.

2.1.3 Operational efficiency and cost optimization

With margins under pressure, operators seek to lower operational expenditures and capital expenditures while handling exponential traffic growth. Autonomous capabilities such as closed-loop automation, self-healing, and proactive assurance reduce manual workloads, accelerate fault resolution, and optimize energy consumption. This shift from reactive to proactive operations yields tangible cost efficiencies and productivity gains.

2.1.4 Customer experience as a strategic imperative

In the digital economy, customer expectations are instantaneous, always-on, and highly contextual. Service degradation or outages quickly translate into churn and reputational loss. AN enables proactive monitoring, predictive assurance, and experience-centric KPIs, shifting the operator's posture from reactive fault management to experience-driven value delivery.

2.1.5 Intent-based operations require optimized resource utilization

Meeting growing customer demand at scale requires networks to manage multiple, often conflicting, intents effectively. Traditional rule-based automation lacks the sophistication to resolve these conflicts dynamically and cannot prioritize intents at scale while maintaining guaranteed Service Level Agreements (SLAs). ANs address this challenge via utility-based decision-making capable of resolving intent conflicts without compromising end-user performance requirements. Utility-based decision-making ensures intelligent trade-offs, balancing latency, throughput, and energy efficiency while autonomously prioritizing critical services (e.g., enterprise 5G slices over lower-priority traffic). This approach guarantees efficient resource allocation and higher customer satisfaction (CSAT) while sustaining competitive business operations.

By embedding utility functions into decision-making engines, Intent-Driven ANs surpass traditional binary rule-based execution. These networks deliver differentiated connectivity that is aligned with both operator business priorities and customer expectations, ensuring adaptable, high-quality service delivery across diverse operational domains.

2.2 Use cases: Intent-based automation

Intent-based automation can be applied across a wide range of scenarios to ensure networks operate in alignment with business and service objectives. For instance, it can dynamically configure and monitor network slices to deliver specific quality of service levels, scaling resources up or down in real-time to meet performance commitments. It can also optimize overall network performance by predicting congestion, adjusting antenna configurations, steering traffic, and balancing loads across sites to maintain seamless coverage and reduce operational costs.

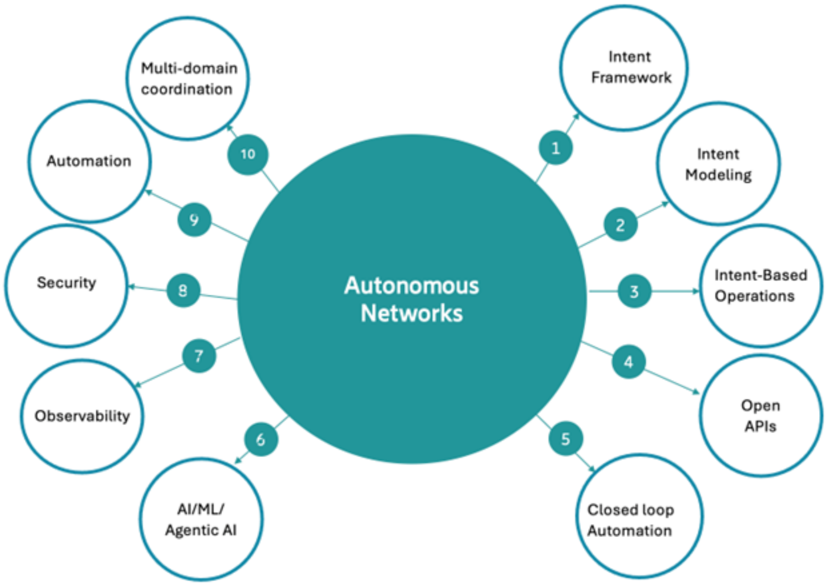
In addition, intent-driven frameworks can improve energy efficiency by applying AI/ML algorithms that analyze traffic patterns and automatically adjust the power consumption of RAN components during off-peak hours. While these illustrate the practical benefits of intent-based automation, the full potential extends much further, with future innovations likely to unlock new and transformative use cases as networks evolve.

3. Building blocks

ANs redefine telecom operations by integrating key building blocks such as intent framework, AI/ML and advanced AI capabilities. High-level intents, including objectives like SLA targets or energy efficiency goals, are dynamically transformed into actionable strategies via Intent Management Functions (IMFs) across business, service, and resource layers. Leveraging AI and Machine Learning, AN ensures adaptive self-configuration, self-optimization, and self-healing, while autonomously resolving conflicts.

Open APIs enable multi-vendor interoperability and support dynamic orchestration, separating business intentions ("why") from technical implementation ("how"). Observability frameworks provide predictive insights for optimizing performance in real-time, while Zero-Trust security safeguards ensure operational resilience and trusted adaptation. Together, these components empower telecom providers to deliver scalable, efficient networks aligned with business goals, capable of dynamically responding to evolving demands in an intelligent and efficient manner. Figure 3.1 and subsequent content provide a non-exhaustive list and dives deeper into these enablers of network autonomy.

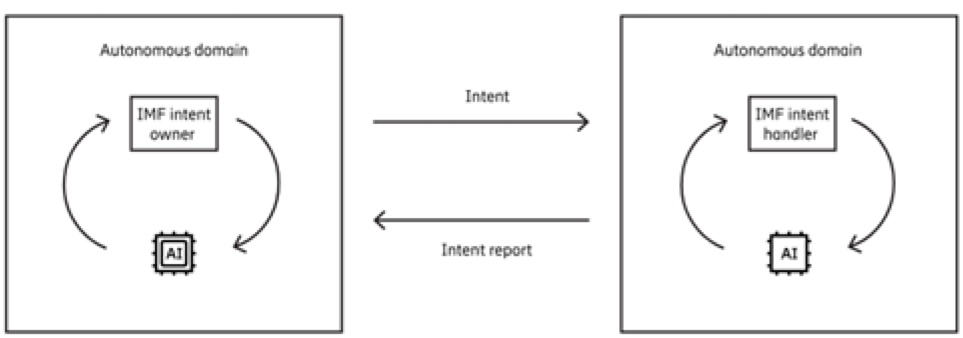
Figure 3.1: Autonomous networks building blocks



3.1 Intent framework

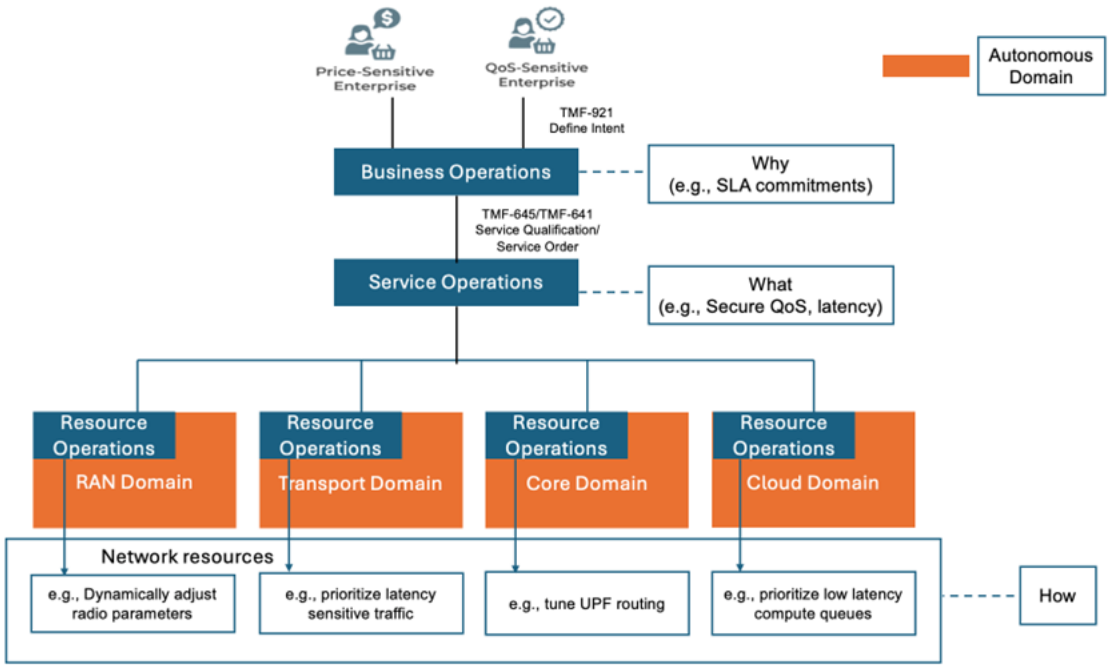
Intents define high-level objectives without prescribing execution details, enabling CSPs to translate business goals—such as Service Level Agreements—into autonomous network actions. Autonomous domains are specialized, self-governing functional areas within a network, defined by specific scopes and exclusive responsibility for managing resources (e.g. RAN, core, transport, etc). Each autonomous domain implements an IMF and exchanges requirements with other domains; an IMF acting as intent owner sends intents to an intent handler realized by another IMF, as shown in Figure 3.2.

Figure 3.2: IMF interaction between AI-driven autonomous domains[7]



This structured cooperation among IMFs ensures that subsystems adapt dynamically to strategic changes while avoiding conflicting resource actions. Features like negotiation, utility-maximizing decisions, and intent reporting facilitate autonomous conflict resolution across multiple operational layers.

Figure 3.3: Representation of Intent Management Framework as prescribed by TM Forum



The intent framework, as shown in Figure 3.3, operates across three interconnected layers, linked via open and standardized interfaces.

The business operations layer accepts product-level intents, such as network connectivity services, and encodes their value through utility functions. These intents are decomposed into customer-facing service-intents for domain-specific execution. The Service Operations layer refines customer facing service-intents are further into resource-facing intents (RFS), driving deployment and configuration across technology domains, including RAN, Core, and transport. The Resource Operations layer is where autonomous domains process RFS intents to activate capabilities such as slicing, quality of service, and dynamic resource allocation, translating initial requirements into domain-specific operational metrics.

Throughout each step of intent decomposition, the original requirements and the utility function are translated into the metrics and constructs available within each autonomous domain. This layered approach ensures seamless intent decomposition while maintaining alignment with overarching business goals.

3.2 Intent modeling

The TMF Intent Ontology (TIO) TR292 provides a standardized, extensible framework for defining intents across domains.[2] Its common models ensure interoperability across IMFs, while extension models support domain-specific needs. Federated ontology graphs enable decentralized integration, fostering multi-vendor collaboration and adaptive responses with minimal human intervention.

3.3 Intent-based operations

Intent-based operations are the foundation of ANs, shifting from manual control to intelligent, goal-driven management. Operators define high-level objectives (intents) which AI systems translate into dynamic network actions for configuration, optimization, and assurance. This model enhances scalability, reliability, and alignment with business outcomes, making it essential to next-generation telecom networks.

3.4 Programmable infrastructure via open APIs

Programmable infrastructure, powered by open APIs, allows network resources to be dynamically controlled, configured, and optimized through standardized, interoperable interfaces. By eliminating manual configurations and proprietary constraints, open APIs enable seamless integration across multi-vendor environments, fostering innovation while supporting dynamic orchestration and resource scaling. They facilitate intent lifecycle management, where AI-driven systems autonomously execute optimization and adaptation tasks. The separation of “why” (intent) from “how” (vendor-specific execution) is central to ensuring interoperability and achieving scalable autonomy in modern networks.

3.5 Closed-loop automation

Closed-loop automation drives autonomous network operations by continuously monitoring performance, analyzing data in real-time, and dynamically adjusting configurations to achieve specified objectives without human intervention. Enabled by AI and ML, this process transitions network management from reactive to predictive control, enhancing adaptability, scalability, and alignment with business priorities. Operating within a self-correcting cycle (Observe → Analyze → Decide → Act), it optimizes resource allocation, ensures reliability through self-healing, and integrates proactive intelligence to align network behavior seamlessly with operational goals like latency and throughput.

3.6 Artificial intelligence/machine learning (AI/ML)

AI and ML form the foundation of ANs, enabling them to operate efficiently and adapt to dynamic demands. While AI provides reasoning and decision-making, ML drives continuous improvement through data. Large Language Models (LLMs) extend these capabilities by interpreting intents, analyzing telemetry, and recommending actions. Further, **telco-grade LLMs**—fine-tuned with domain knowledge—offer the precision, reliability, and explainability required for mission-critical environments. Together, AI/ML and telco-grade LLMs create the cognitive core of intent-driven ANs, shifting operations from reactive automation to proactive, adaptive, and monetizable connectivity.

3.7 Agentic AI

Agentic AI marks a major leap toward ANs by deploying modular AI agents that independently pursue specific goals while collaborating to optimize performance. These agents make decisions, adapt in real-time, and reduce reliance on human intervention—whether improving service availability or operational efficiency. The **Model Context Protocol (MCP)** underpins this architecture, providing the interoperability, governance, and coordination needed for multi-agent systems to function across domains and vendors. By aligning with TM Forum standards, MCP ensures transparent collaboration and scalability, making Agentic AI a cornerstone of next-generation telecom autonomy.

3.8 Observability

Observability ensures continuous visibility into system states by analyzing outputs, which is essential for managing increasingly complex and resource-intensive networks. It provides operators with real-time intelligence to monitor service performance and network conditions comprehensively. As a critical building block for ANs, observability feeds AI models rich, contextualized data, enabling predictive operations, automated troubleshooting, and proactive optimization. This framework ensures AI-driven networks maintain service quality, resilience, and independence through trustworthy, granular insights.

3.9 Security

Security ensures AI-driven ANs operate safely by mitigating threats and vulnerabilities. Key components like Zero-Trust authentication, AI model integrity protection, and cryptographic domain boundaries safeguard systems while enabling secure automation. With tools such as continuous monitoring, risk-adaptive policies, and identity management, networks can make independent decisions and prevent unauthorized access. These foundational frameworks guarantee operational integrity, compliance, and resilience in fully autonomous environments.

3.10 Intent-driven automation

Intent-driven automation (IDA) extends CI/CD pipelines to manage high-level intents rather than low-level code. Intents are version-controlled as the source of truth, validated for syntax and policy compliance, and deployed through automated orchestration. Continuous assurance (CA) closes the loop by monitoring live systems and triggering remediation, while AI/ML predicts and prevents degradations.

This enables ANs to dynamically translate intent into action with reliability, compliance, and efficiency—minimizing human oversight.

3.11 Multi-domain coordination

Intent-based network management (IBN) enables seamless multi-domain coordination by translating business intents into service- and resource-level actions across autonomous domains such as RAN, transport, core, and cloud. It ensures end-to-end alignment through unified translation, allows domains to operate independently yet collaboratively, and leverages adaptive orchestration to respond dynamically to changing demands. Real-time telemetry sharing drives immediate adjustments, fostering interoperability, agility, and scalability essential for ANs in complex environments.

4. Telco industry landscape

The acceleration toward intent-based ANs has evolved from conceptual frameworks into tangible global initiatives, reflecting concerted momentum across industry, standards bodies, and research communities. These efforts aim to define intent models and procedures for network, core, and RAN management, paving the way for simplified operations and higher levels of automation in the network.

4.1 Standards-led transformation

4.1.1 TM Forum

The TM Forum's ANs framework has become a de facto standard for operators to benchmark their maturity and guide investment, with defined levels progressing from basic assistance to fully autonomous operations. For any operator trying to assess their current maturity level, they need to use the TM Forum Autonomous Network Level Evaluations Tool (ANLET) and a validation service (ANLAV).[3] ANLET was introduced in 2024 to help CSPs benchmark their network automation maturity and track progress toward achieving Level 4 in various network scenarios whereas ANLAV was introduced in 2025 DTW ignite which lets CSPs independently validate their autonomous network levels. Figure below gives a view of the maturity levels prescribed by TM Forum with the highlighted level indicating the current industry vision.

Through initiatives such as open APIs, intent-based management models, and catalyst programs, TM Forum accelerates the adoption of autonomous network principles, fosters innovation, and promotes a unified approach to achieving fully autonomous operations in telecommunications.

4.1.2 O-RAN Alliance

O-RAN Alliance has studied intent-driven management which introduces the exposure of the SMO services for RAN management in a simplified form via intents. This includes various interaction within the small and medium-sized enterprise (SME) management loops (closed, open), serving as a simplified method for exposure of RAN management services to other internal or external management systems.[5] The study also addresses resolution of conflict in intents and provides recommendations on requirements, architectural considerations, and interface designs to support scalable and flexible intent-driven RAN management, setting the foundation for subsequent feature development phases.

4.1.3 3GPP

3GPP defines intent-driven management as a service-based approach where a consumer specifies high-level, unambiguous intents that describe what needs to be achieved in network or service management, without dictating how to achieve it. 3GPP intent-driven management is specified in Technical Specification (TS) 28.312.[6] There are continued efforts in 3GPP to leverage intent-driven management to improve network automation and service delivery.

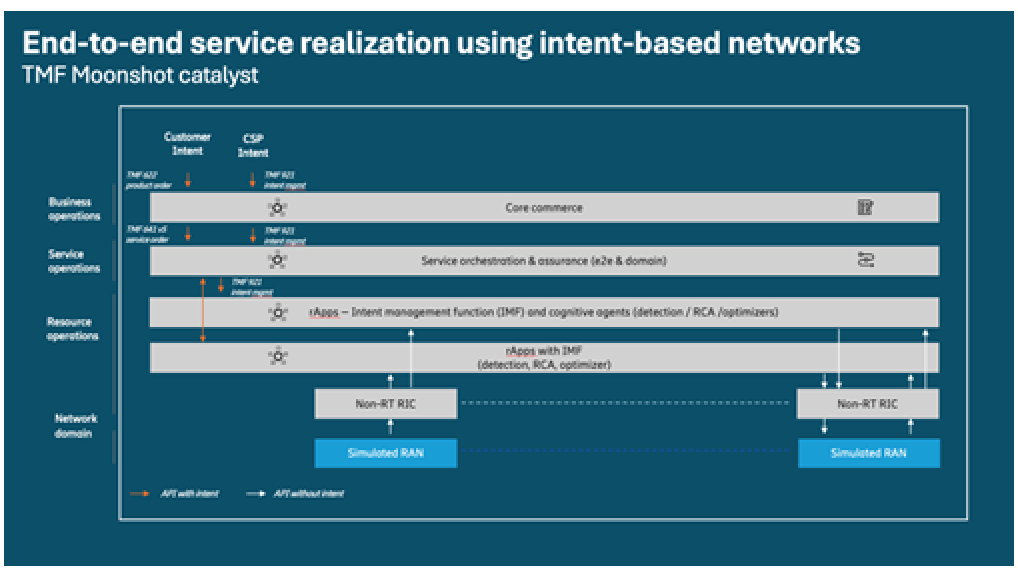
4.2 Vendor ecosystem

The telco vendor ecosystem is advancing ANs by embedding intent-based capabilities into BSS/OSS and RAN domains. In BSS/OSS, vendors are modernizing with open APIs, microservices, data fabrics, and Agentic workflows leveraging telco-grade LLMs to enable intent translation, and service agility in customer experience, billing, and service fulfillment and assurance workflows. Within the O-RAN RIC ecosystem, the SMO and its rApps provide modular, programmable functions that translate business and operational intents into policies for RAN optimization, demonstrating how autonomy can scale incrementally across domains while maintaining interoperability.

Collaboration with standards bodies like TM Forum is central to this progress. Projects such as the Moonshot Catalyst on End-to-End Service Realization showcased how TM Forum open APIs can integrate across business, SMO, and rApp layers in a live 5G environment, demonstrating seamless integration of TM Forum APIs for intent-based management across multiple layers, encompassing business objectives, SMO, and rApp deployment.[7] It

Figure 4.1: Autonomous network levels[4]

Figure: 4.2 TMF Moonshot catalyst project M25.0.799[8]



facilitated the fulfillment of customer intent for premium service delivery while aligning with operator objectives of energy efficiency.

Overall, these developments are expected to enhance CSAT, lower operational costs, and maximize the efficiency of investment in network assets.

4.3 Operator journey

4.3.1 Global

Over 30 CSPs have undergone AN Level evaluations (ANLET), with TM Forum introducing ANLAV to validate results against high-value scenarios, some of which are mentioned below. These initiatives reflect a clear global trend: tasks once managed manually are increasingly automated through intelligent systems, signaling advancing maturity of ANs across the industry. Progress has been demonstrated across diverse domains and use cases, including customer experience management, predictive operations, network optimization, resource planning, and real-time visualization. These initiatives highlight a broader global trend toward higher levels of autonomy, where tasks once managed manually are increasingly supported by intelligent systems. Collectively, these developments illustrate the advancing maturity of autonomous network capabilities across the industry.

4.3.2 North America

In North America, operators are advancing toward intent-based ANs with phased strategies integrating AI, cloud-native architectures, and network slicing in their networks as well as actively exploring advanced AI capabilities such as Generative AI and Agentic AI. While still in early adoption phases, only 33 percent of regional operators have reached AN maturity levels 2/3 in specific high-value use cases, aiming to achieve level 4 within the next two to three years.[9] Intent-based automation is already transforming domains like fault management, energy resource optimization, and network change management, aligned with TM Forum-prescribed maturity levels. Additional focus is on predictive maintenance and resource optimization.

North American operators are actively engaging with TM Forum through initiatives like the autonomous networks manifesto and Moonshot catalyst projects, demonstrating a strong commitment to innovation and advancing network autonomy. These collaborations underscore the growing momentum toward transforming telecom networks into adaptive, efficient, and self-governing systems.

5. Challenges to address

Operators are transitioning from pilot projects to production, yet progress often stalls due to the absence of clear priorities, end-to-end standardization, and measurable business value. Key barriers hindering adoption include but are not limited to:

Legacy systems and migration complexity: Operators face challenges in modernizing entrenched monolithic systems and breaking down data silos. Decisions regarding "build versus buy" approaches for AI-driven operations add to the complexity.

Model/Agent Observability and Governance: Establishing confidence in AI models and autonomous agent decisions, alongside implementing robust observability across diverse autonomous domains and multi-vendor environments, is crucial for ensuring predictable, reliable, and trustworthy network behavior.

Competing Investments: Operators must strategically prioritize scenarios with tangible Return on Investment (ROI) rather than focus solely on technologically impressive solutions. Choosing initiatives with clear business outcomes is essential to sustain progress.

Operational Readiness: Achieving AN success requires robust infrastructure prepared for autonomous functions, integrating seamlessly across on-premises, hybrid, and cloud-native environments. Legacy systems often demand significant upgrades or replacements, adding complexity to interoperability efforts.

Security: AI-driven automation increases vulnerability to cyber threats due to dynamic decision-making processes and reduced human oversight. Multi-vendor environments and open frameworks like O-RAN amplify security risks, necessitating adaptive measures and robust protection protocols.

Data Management: Effective management focuses on resolving semantic inconsistencies across telecom domains and vendors while unifying telemetry pipelines for real-time, secure decision-making. Investments in data tools are essential in overcoming silos and facilitating actionable insights.

Trust: Operators are hesitant to adopt widespread AI mechanisms due to uncertainties about reliability, scalability, and fault-free outcomes. Transparency initiatives, such as TM Forum frameworks, are crucial in building confidence through iterative improvements.

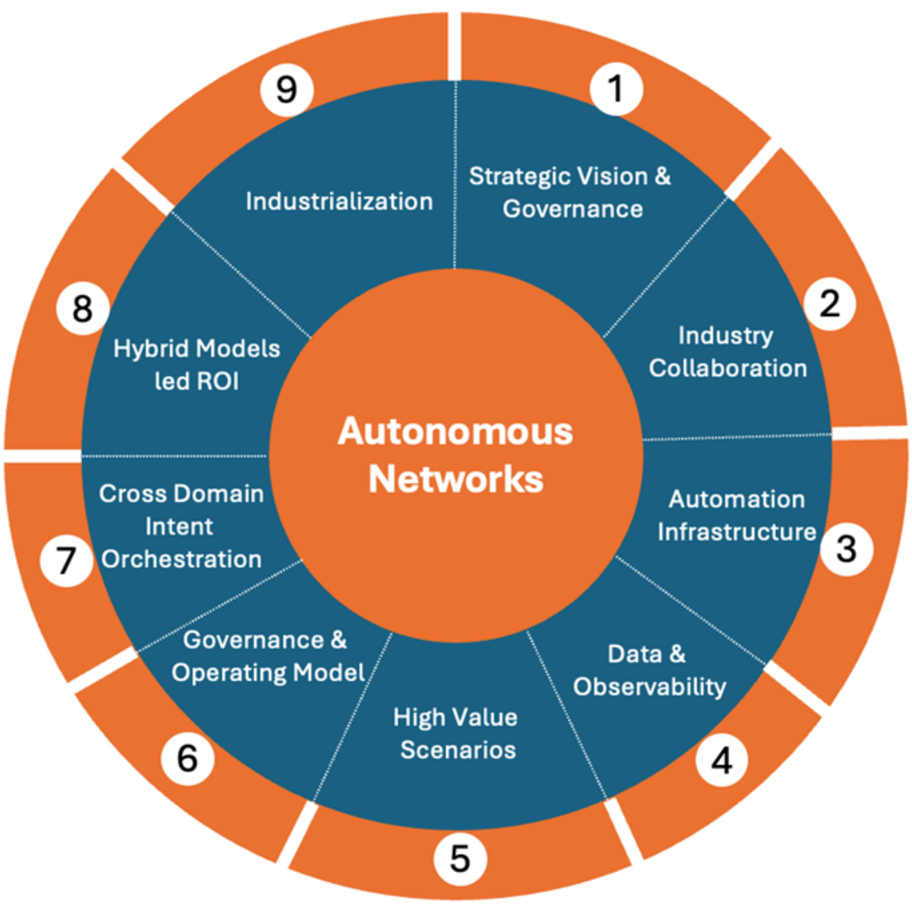
Investment: Full automation demands significant capital for integrating systems like digital twins, intent management APIs, and machine reasoning engines. Strategic investments are key to achieving scalable and intent-driven operations.

Despite these challenges, industry eagerness to adopt ANs persists. Operators are focused on navigating barriers to unlock the transformative potential of AN frameworks, combining phased advancements with collaborative strategies to ensure scalable, secure, and efficient deployment.

6. Roadmap to implementation

Advancing from traditional operations to autonomous capabilities requires a phased, multi-year journey that balances ambition with risk management and adapts to each CSP's priorities and context. Achieving Level 4+ autonomy cannot be delivered by a single technology or restructuring; it demands incremental capability-building with measurable value at every stage. Strategies vary—some operators may focus on domain-specific autonomy (e.g., RAN) before expanding to core and transport, while others pursue cross-domain orchestration from the outset. Adoption paths depend on infrastructure maturity, organizational readiness, competitive positioning, and regulatory requirements, but all must preserve service excellence while progressing toward scalable autonomy.

Figure 6.1: Roadmap to Implementation



The following steps provided a non-exhaustive guideline for methodical adoption of autonomous capabilities in today's complex networks.

6.1 Define strategic vision and governance

Drive network transformation through measurable business outcomes—such as faster Mean Time To Resolution (MTTR), improved CSAT scores, greater energy efficiency, and new service enablement (e.g., 5G slicing)—while securing executive alignment on ROI-focused priorities.

6.2 Build collaborative industry alignment via standards adoption

Adopt standards-led frameworks such as the one from TM Forum in designing autonomous network systems. Participate in TM Forum Initiatives such as AN maturity assessment, Moonshot Catalysts, Open Digital Framework projects, and MCP evolution for promoting cross-domain, multi-vendor interoperability.

6.3 Invest in automation-ready infrastructure

Focus investments on upgrading network architecture to automation-ready standards, prioritizing modernization over incremental improvements to overcome technical debt. Promote use of open APIs to solve interoperability issues between domains, vendors, technologies and infrastructures.

6.4 Establish data and observability foundations

Work towards implementing a federated data fabric. Harmonize domain-specific datasets (RAN, transport, core, OSS/BSS, customer contexts) via TM Forum ontologies. Enable observability dashboards across domains powered by open APIs to ensure network-wide lifecycle transparency.

6.5 Identify and implement high-value scenarios

This phase focuses on identifying and implementing high-value, low-risk automation use cases that deliver immediate business impact while building organizational confidence and scalability foundations. Operators prioritize scenarios aligned with key objectives—such as reducing churn, optimizing resources, or improving productivity—with examples including proactive customer experience monitoring, network resource optimization, and workflow simplification. TM Forum's catalog of 20 High-Value Scenarios provides guidance for selecting the most impactful use cases.

6.6 Establish governance and operating models

Focus on establishing robust governance frameworks to manage autonomous network operations at scale. This includes defining decision-making authority, risk tolerance, and operational boundaries for autonomous systems. It provides the organizational structures and processes necessary for responsible deployment, ensuring operational control, regulatory compliance, and measurable business value.

6.7 Build capabilities for intent translation and cross-domain orchestration

This phase focuses on transforming high-level business objectives into network actions through intent-based frameworks. It involves establishing cognitive systems that translate business intent into coordinated configurations across domains (RAN, core, transport, and service) and deploying cross-domain orchestration frameworks to ensure seamless optimization across multiple layers.

6.8 Maximize ROI from targeted AI models

Incorporate hybrid AI models. To balance cost vs efficiency, employ small domain-specific models for use cases like monitoring and anomaly detection and reserve LLMs for complex reasoning, root-cause analysis, and customer experience (CX) personalization.

6.9 Accelerating the journey to production-grade ANs

While standards bodies such as TM Forum prescribe a structured framework for autonomous networks including AN levels, open APIs, and ODA, industrialization of network autonomy requires establishment of industry consortiums that bring together operators, vendors, and standards bodies to co-develop interoperable methodologies, align intent frameworks, and deliver multi-domain, multi-vendor, production-grade implementations.

Conclusion

The transition toward intent-driven autonomous networks marks one of the most pivotal evolutions in telecommunications history. Success will depend on decisive leadership, industry-wide collaboration, and commitment to advancing from isolated pilots to production-grade, interoperable systems. While global efforts are valuable, it is North America that must set the pace—not only for its own competitiveness, but to define the frameworks and practices that will guide worldwide adoption.

North America's legacy of wireless leadership—from Long-Term Evolution (LTE) to 5G—provides a strong precedent. The region's operators, vendors, and standards contributors have consistently shaped the course of mobile technology, and the same opportunity exists today with ANs. By acting as the proving ground for high-value use cases, North America can demonstrate the business outcomes, governance models, and AI-enabled capabilities that will make intent-based automation scalable and trusted.

This leadership role is not optional—it is strategic. By pioneering ANs, North America can drive digital economy growth, reinforce global competitiveness, and establish a resilient, sustainable, and future-ready telecom infrastructure. The outcome will be more than operational efficiency; it will be a redefinition of telecom as an adaptive, monetizable, and customer-centric platform for innovation.

Takeaway: The journey to network autonomy is phased, but leadership is immediate. North America must seize the moment to pioneer intent-based autonomous networks and, in doing so, shape the future of global connectivity.

Appendix

Acronyms

AI: Artificial Intelligence

AI/ML: Artificial Intelligence/Machine Learning

AN: Autonomous Networks

ANLAV: Autonomous Network Level Assessment Validation

ANLET: Autonomous Network Level Evaluations Tool

BSS: Business Support Systems

CA: Continuous assurance

CI/CD: Continuous integration/Continuous Deployment

CNCF: Cloud-Native Computing Foundation

CSAT: Customer Satisfactions

CSP: Communications service providers

CX: Customer Experience

DTW: Digital Transformation World

IBA: Intent-Based Automation

IBAN: Intent-Based Autonomous Networks

IBN: Intent-Based Networking

IDA: Intent-driven automation

IMF: Intent Management Function

IoT: Internet of Things

KPI: Key Performance Indicators

MCP: Model Context Protocol

MTTR: Mean Time to Resolution

ODA: Open Digital Architecture

OSS: Operations Support Systems

QoS: Quality of Service

RAN: Radio Access Networks

RIC: RAN Intelligent Controller

SLA: Service Level Agreements

SME: Small and Medium-sized Enterprise

SMO: Service Management and Orchestration

SON: Self-Organizing Networks

TMF: TM Forum

TIO: TMF Intent Ontology

TS: Technical Specification

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6 Directory Listing /ftp/Specs/archive/28_series/28.312

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8 End-to-end service realization using intent-based networks

9 Networks with intelligence: Why and how the telecom sector should accelerate its autonomous networks journey

Acknowledgments

5G Americas' Mission Statement: To be the trusted voice in North America's wireless industry, fostering collaboration, driving thought leadership, and advancing next-generation mobile technologies.

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